

MR_IL6R_CRC v44 validation, Figure 1, and reference paper package

Output time: 2026-05-16 19:05 北京时间

Executive conclusion

Prepared independent validation add-on, Figure 1 full-width exports, Yanggang Hong paper pack, and all cited reference manifest. Repository URL replacement is paused.

Independent validation

Run the provided GEO scripts before final submission to reduce the review risk of a computational-only study.

Figure 1

The package includes a 2100-px-wide 300-dpi PNG, PDF/SVG wrappers, and a clean native-vector redraw.

Reference packaging

Locally available PDFs were packaged. Remaining references are listed in DOI manifests and require legal open-access/manual download.

External validation cohort candidates

cohort	priority	suggested use	reference_url
GSE39582	High	Primary external validation: IL6R expression, survival/Cox if phenotype fields available, immune/inflammatory association	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE39582
GSE17536	High	External survival validation and directionality check for IL6R expression	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE17536
GSE17537	Medium	Sensitivity validation for OS/metastasis endpoints	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE17537
GSE14333	Medium	Additional expression/prognosis validation if phenotype fields support endpoint extraction	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE14333

Figure 1 export audit

file	type	width_px	height_px	dpi	note
Figure_1_workflow_fullwidth_300dpi_raster_v44_20260516_1905.png	raster_png	2100	1575	300	Upscaled from v40 PNG; improves file dimension but not true native vector detail.
Figure_1_workflow	raster_wrapped_p	2100	1575	embedded 300	PDF wrapper

w_fullwidth_300dpi_raster_wrapped_v44_20260516_1905.pdf	df			logical	around 300dpi raster image.
Figure_1_workflow_fullwidth_300dpi_raster_wrapped_v44_20260516_1905.svg	raster_wrapped_svg	2100	1575	300 logical	SVG wrapper around base64 raster image; not true editable vector.
Figure_1_workflow_vector_redraw_clean_v44_20260516_1905.svg	native_vector_svg	2400	1600	vector	Clean vector-style redraw; not identical to image2 icons but fully scalable.
Figure_1_workflow_vector_redraw_clean_v44_20260516_1905.pdf	native_vector_pdf	2400	1600	vector	PDF generated from vector SVG.

Yanggang Hong core reference papers

reference_role	doi	title	fixed_status
核心蓝图 A	10.1016/j.ecoenv.2025.118709	Integrative causal and single-cell analyses reveal genes responsive to endocrine disruptors driving human male infertility	included
核心蓝图 B	10.1016/j.ecoenv.2025.118126	Prioritizing endocrine-disrupting chemicals targeting systemic lupus erythematosus genes via Mendelian randomization and colocalization analyses	included
核心蓝图 C	10.1016/j.reprotox.2024.108749	di-ethylhexyl phthalate with network toxicology and molecular docking strategy	included
核心蓝图 D	10.1016/j.arr.2025.102835	Mapping the immune-genetic architecture of aging: A single-cell causal framework for biomarker discovery and therapeutic targeting	included
核心蓝图 E	10.1186/s10194-025-02132-x	The Journal of Headache and Pain Open Access RESEARCH Dissecting immune cell-specific genetics in migraine: a multi-omics framework for target discovery and therapeutic prioritization	included
机制验证 A	10.1016/j.ecoenv.2024.117352	Environmental triggers and future risk of developing autoimmune diseases: Molecular mechanism and network toxicology analysis of bisphenol A	included
机制验证 B	10.1016/j.ecoenv.2025.117878	Analysis of toxicity and mechanisms of busulfan in non-obstructive azoospermia: A genomic and toxicological approach integrating molecular docking, single-cell sequencing, and experimentation in vivo	included

癌症整合 A	10.1186/s12967-025-06517-z	Decoding per? and polyfluoroalkyl substances (PFAS) in hepatocellular carcinoma: a multi?omics and computational toxicology approach	included
癌症整合 B	10.1007/s10142-025-01616-y	Analyzing the potential targets and mechanism of per- and polyfluoroalkyl substances (PFAS) on breast cancer by integrating network toxicology, single-cell sequencing, spatial transcriptomics, and molecular simulation	included
免疫遗传 A	10.1093/rheumatology/keaf606	and colocalization reveal cell-specific mechanisms in systemic lupus erythematosus Yanggang Hong 1 2 1,?	included
免疫遗传 B	10.1096/fj.202503032R	Single--Cell Transcriptome--Wide Mendelian Randomization and Colocalization Analyses Uncover Cell--Specific Mechanisms in Epigenetic Age Acceleration Yanggang Hong1,2	included
转化靶点 A	10.1038/s41531-025-01148-z	Published in partnership with the Parkinson's Foundation Peripheral immune cell-speci?c genes in Parkinson's disease uncovered by multiomics with therapeutic implications Check for updates 1,2 1234567890());,; 1234567890());,;	included

Reference PDF availability summary

metric	value
total_v43_references	36
local_pdfs_included	5
not_available_locally_or_needs_OA_download	31